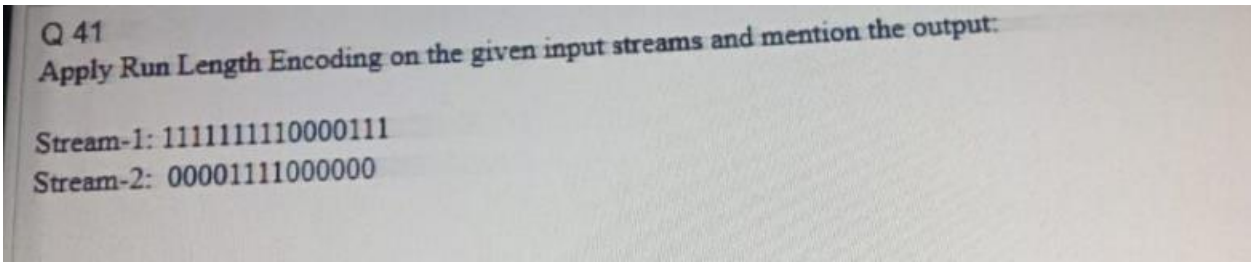
Cs614 current paper solved



🧮 Input Stream-1: `11111111100000111`

🔍 Step-by-step breakdown:

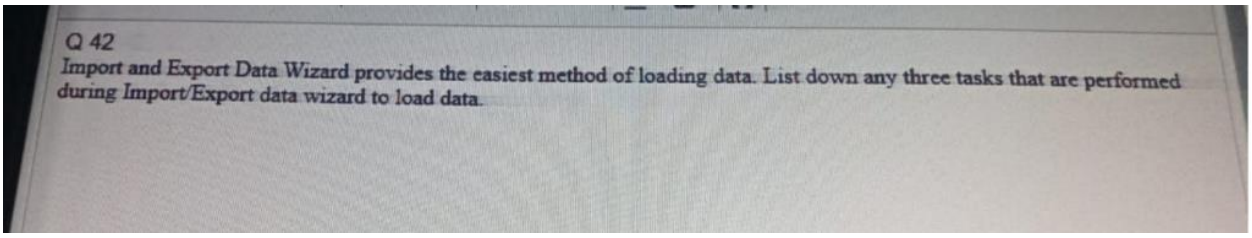
- `1` repeated **9** times → `91`
- `0` repeated **5** times → `50`
- `1` repeated **3** times → `31`

🧮 Stream-2: `00001111000000`

Break it down:

- `0` repeated 4 times → `40`
- `1` repeated 4 times → `41`
- `0` repeated 6 times → `60`

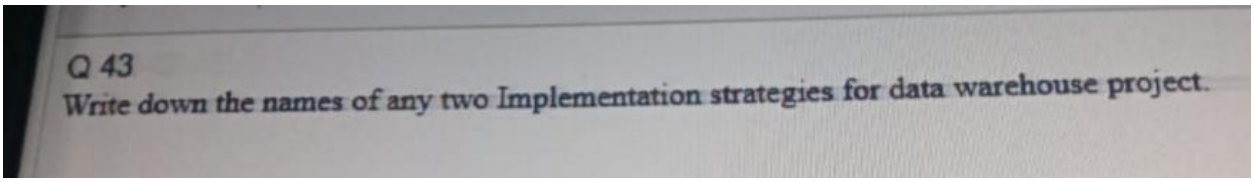
🔧 RLE Output for Stream-2: `40 41 60`



Q 42

Import and Export Data Wizard provides the easiest method of loading data. List down any three tasks that are performed during Import/Export data wizard to load data.

- Establish connection through source / destination systems
- Creates similar table in SQL Server
- Extracts data from text files
- Apply very limited basic transformations if required
- Loads data into SQL Server table



Q 43

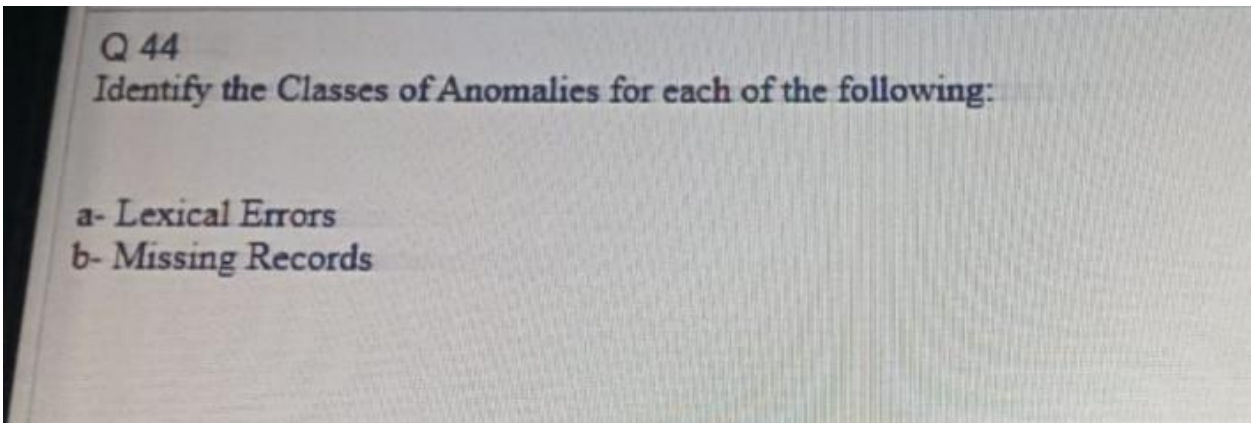
Write down the names of any two Implementation strategies for data warehouse project.

warehouse development methods can fall within three bas
driven and user-driven.

Implementation strategies

- Top down approach
- Bottom Up approach

Important



Q 44

Identify the Classes of Anomalies for each of the following:

a- Lexical Errors
b- Missing Records

- Syntactically Dirty Data
  - Lexical Errors
- Coverage Anomalies
  - Missing Attributes
  - Missing Records

Question 45

Data Warehousing (CS614)

Question: 45 (Marks: 3) Atten

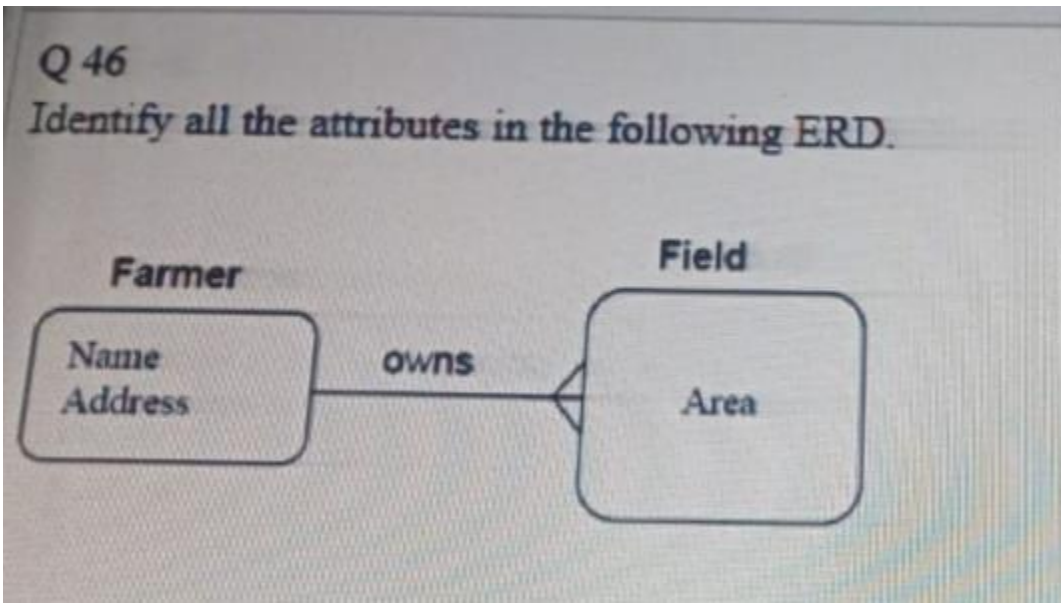
From Dense and Sparse Indexes, identify the correct index for each of the following:

1. If fits in the memory, costs only one disk I/O access to locate a record given a key.
2. Uses less space, but the block has to be searched, even for unsuccessful searches.

Answer:

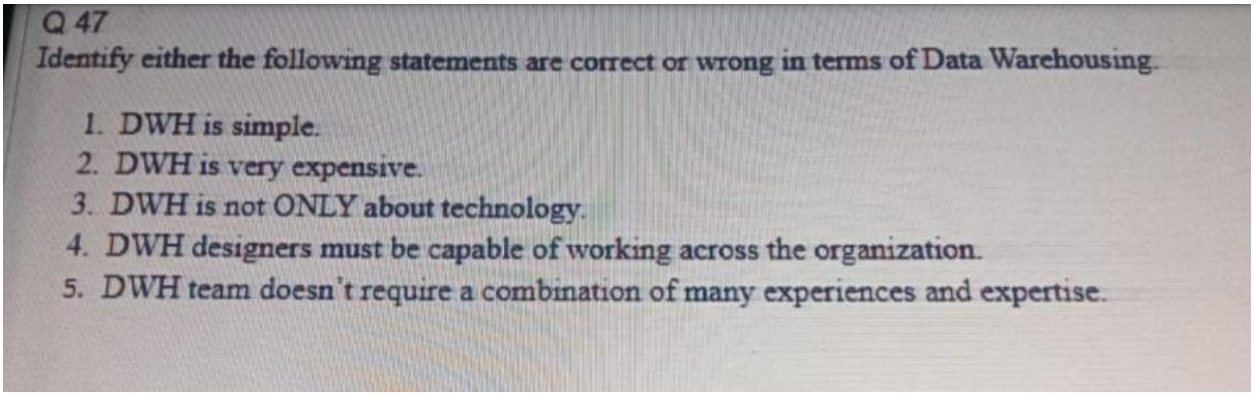
1. Dense Index

2. Sparse Index



✅ Attributes in the ERD:

- **Farmer:** `Name`, `Address`
- **Field:** `Area`

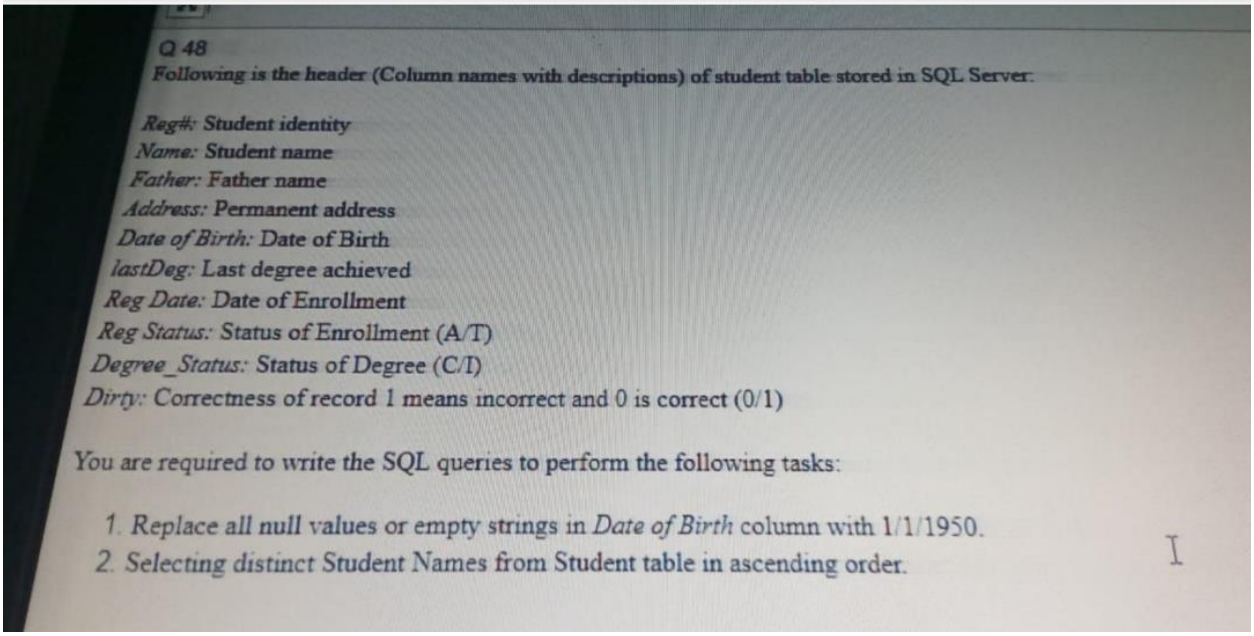


Q 47

Identify either the following statements are correct or wrong in terms of Data Warehousing.

1. DWH is simple.
2. DWH is very expensive.
3. DWH is not ONLY about technology.
4. DWH designers must be capable of working across the organization.
5. DWH team doesn't require a combination of many experiences and expertise.

1. **DWH is simple → ❌ Wrong**
2. **DWH is very expensive → ✅ Correct**
3. **DWH is not ONLY about technology → ✅ Correct**
4. **DWH designers must be capable of working across the organization → ✅ Correct**
5. **DWH team doesn't require a combination of many experiences and expertise → ❌ Wrong**



Q 48

Following is the header (Column names with descriptions) of student table stored in SQL Server:

Reg#: Student identity
Name: Student name
Father: Father name
Address: Permanent address
Date of Birth: Date of Birth
lastDeg: Last degree achieved
Reg Date: Date of Enrollment
Reg Status: Status of Enrollment (A/T)
Degree_Status: Status of Degree (C/I)
Dirty: Correctness of record 1 means incorrect and 0 is correct (0/1)

You are required to write the SQL queries to perform the following tasks:

1. Replace all null values or empty strings in *Date of Birth* column with 1/1/1950.
2. Selecting distinct Student Names from Student table in ascending order.

✅ Task 1: Replace null or empty strings in `Date of Birth` with `1/1/1950`

Sql — Copy

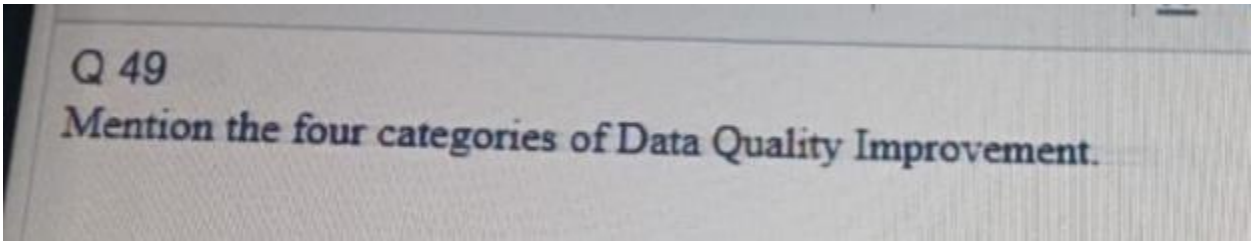
```sql
UPDATE Student
SET [Date of Birth] = '1950-01-01'
WHERE [Date of Birth] IS NULL OR [Date of Birth] = '';
```

✅ Task 2: Select distinct student names in ascending order

Sql — Copy

```sql
SELECT DISTINCT Name
FROM Student
ORDER BY Name ASC;
```

Let me know if you want these queries explained or adapted for a different SQL dialect.



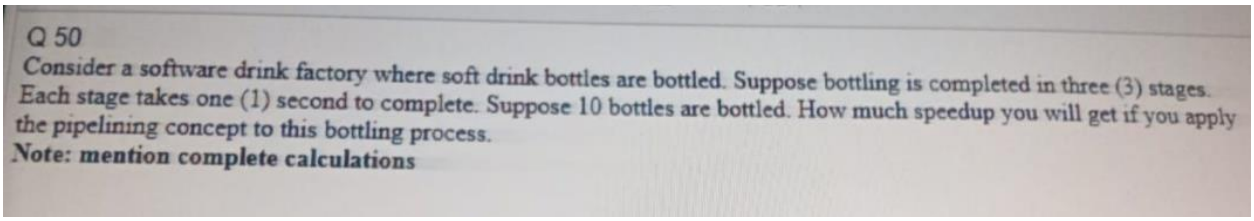
Q 49

Mention the four categories of Data Quality Improvement.

23.4 How to improve Data Quality?

The four categories of Data Quality Improvement

- Process
- System
- Policy & Procedure
- Data Design



Q 50

Consider a software drink factory where soft drink bottles are bottled. Suppose bottling is completed in three (3) stages. Each stage takes one (1) second to complete. Suppose 10 bottles are bottled. How much speedup you will get if you apply the pipelining concept to this bottling process.

Note: mention complete calculations

🧮 Without Pipelining (Sequential Execution)

Each bottle takes 3 seconds (1 second per stage), and there are 10 bottles:

- **Total time** = 10 bottles × 3 seconds = **30 seconds**

⚙️ With Pipelining

In pipelining, after the first bottle completes Stage 1, the second bottle enters Stage 1 while the first moves to Stage 2, and so on.

- **Time to fill pipeline** = 3 seconds (first bottle completes all 3 stages)
- **Time for remaining bottles** = 1 second per bottle × (10 - 1) = 9 seconds
- **Total time** = 3 + 9 = **12 seconds**

↓

Speedup formula

Time without pipeline / time with pipeline = 30/12=2.5x